A proof of the conjectured recurrence for OEIS A191790

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Abstract

OEIS A191790 carries a conjectured linear recurrence with polynomial coefficients, of order 5, contributed by R. J. Mathar. We prove it. The generating function of the entry is algebraic, and the recurrence is equivalent to the vanishing of an explicit differential operator applied to it: writing $\theta = x\,\frac{d}{dx}$, the sequence $b(n) = \sum_i p_i(n)a(n-i)$ has generating function $B(x) = \sum_i x^i\,(p_i(\theta+i)A)(x)$, so the conjecture holds for all $n > d$ exactly when B is a polynomial of degree d. Since A lies in a quadratic extension of $\mathbb{Q}(x)$, which is closed under θ, the residual B can be computed in closed form. Here it equals $3x^2$, a polynomial of degree 2, which proves the recurrence for every $n > 2$.

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1 The sequence and the conjecture

OEIS A191790 is "Number of base pyramids in all length n left factors of Dyck paths". It has offset 0 and begins

$$a(0), \ldots, a(7) \;=\; 0, 0, 1, 1, 4, 5, 14, 20,\; \ldots$$

The entry gives the generating function

$$A(x) \;=\; \sum_{n\geq 0} a(n)\,x^n \;=\; -\frac{\left(\sqrt{1-4x^2}-1\right)^2}{2x\,(x^2-1)\left(2x+\sqrt{1-4x^2}-1\right)}, \tag{1}$$

and it carries the following comment:

> **Conjecture:** (n+1)*a(n) +(-n-1)*a(n-1) +5*(-n+1)*a(n-2) +(5*n-11)*a(n-3) +2*(2*n-3)*a(n-4) +4*(-n+3)*a(n-5)=0. - *R. J. Mathar*, Jun 14 2016

As of the "Last modified" line on the live entry (2017-03-28, revision 18) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^{5} p_i(n)\,a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = n{+}1, \qquad p_1(n) = -n{-}1, \qquad p_2(n) = -5\,(n-1)\,, \qquad p_3(n) = 5n{-}11, \qquad p_4(n) = 2\,(2n-3)\,, \qquad p_5$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n\geq 0} a(n)x^n$ be the generating function of a sequence, and let $\theta = x\,\dfrac{d}{dx}$, so that θ acts on x^m as multiplication by m. For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m)a(m)x^m$.

Lemma 1. *Let $p_0, \dots, p_r$ be polynomials and put $b(n) = \sum_{i=0}^{r} p_i(n)\, a(n-i)$, with the convention $a(m) = 0$ for $m < 0$. Then*

$$B(x) := \sum_{n \geq 0} b(n) x^n = \sum_{i=0}^{r} x^i \left(p_i(\theta + i) A\right)(x).$$

Proof. Fix i and substitute $m = n - i$:

$$\sum_{n \geq 0} p_i(n) a(n-i) x^n = \sum_{m \geq 0} p_i(m+i) a(m) x^{m+i} = x^i \sum_{m \geq 0} p_i(m+i) a(m) x^m = x^i \big(p_i(\theta+i)A\big)(x),$$

the last step because $p_i(\theta + i)$ multiplies the coefficient of x^m by $p_i(m+i)$. Summing over i gives the claim. $\square$

Corollary 2. *With the notation of Lemma 1, the recurrence $\sum_i p_i(n) a(n-i) = 0$ holds for every $n > d$ if and only if B is a polynomial of degree at most d.*

Proof. $b(n)$ is the coefficient of x^n in B, so $b(n) = 0$ for all $n > d$ says exactly that B has no terms above x^d. $\square$

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A, and one only has to recognise it.

Lemma 3. *Let $D \in \mathbb{Q}(x)$ and let $K = \mathbb{Q}(x)\big[\sqrt{D}\big]$. Then K is closed under θ; explicitly,*

$$\theta\big(u + v\sqrt{D}\big) = x\, u' + \Big(x\, v' + \frac{x\, v\, D'}{2D}\Big)\sqrt{D}, \qquad u, v \in \mathbb{Q}(x).$$

Proof. Differentiate: $(\sqrt{D})' = D'/(2\sqrt{D}) = \big(D'/(2D)\big)\sqrt{D}$, and multiply by x. Both components remain in $\mathbb{Q}(x)$. $\square$

Consequently, if $A \in K$ then every $p_i(\theta + i)A$ is in K, and so is B. Writing $B = P + Q\sqrt{D}$ with $P, Q \in \mathbb{Q}(x)$, the test of Corollary 2 becomes: B is a polynomial precisely when $Q = 0$ and P has constant denominator. Both are decided by exact cancellation of rational functions.

3 The computation for A191790

The generating function (1) lies in $K = \mathbb{Q}(x)[\sqrt{D}]$ with

$$D \;=\; -\,(2x-1)\,(2x+1)\,.$$

Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K by Lemma 3, the $\sqrt{D}$ component of B cancels identically and the rational part collapses to

$$B(x) \;=\; 3x^2,$$

a polynomial of degree 2.

Theorem 4. *The conjectured recurrence*

$$(n+1)\, a(n) + (-n-1)\, a(n-1) - 5\,(n-1)\; a(n-2) + (5n-11)\, a(n-3) + 2\,(2n-3)\; a(n-4) - 4\,(n-3)\; a(n-5) \;=\; 0$$

holds for every $n > 2$.

Proof. Immediate from Corollary 2 and the displayed value of B. $\square$

Remark 5. The polynomial B is not an error term: it records the boundary of the recurrence. For $n \leq 2$ some of the shifted terms $a(n-i)$ fall outside the range of the sequence, and B collects exactly those contributions. Above that range the relation is exact.

4 Verification

Every number above is an output of a script written separately from this note, and two independent checks were run.

First, the generating function was checked against the entry rather than assumed: the Taylor coefficients of (1) were expanded and compared term by term with the DATA section of A191790, agreeing on all 13 terms compared.

Second, the recurrence itself was evaluated directly on the published terms, in exact integer arithmetic and without reference to the generating function: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 33 values of n from $n = 5$ upward. This is what Corollary 2 predicts from the degree of B, and it is a genuinely different computation from the symbolic one: it never forms B, never differentiates, and uses only integers.

The symbolic step is exact throughout. The residual is obtained by rational-function cancellation in K, not by series truncation, so the identity $B = 3x^2$ is an identity of functions and the theorem holds for all $n > 2$ simultaneously.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A191790.

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[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999, Chapter 6 (algebraic and D-finite generating functions).